

CRITICAL RESEARCH REVIEW ARTICLE

[THE ROLE OF DIGITAL FORENSICS IN CYBERCRIME INVESTIGATIONS]

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Abstract

Write a concise summary of the entire review article. Write a concise summary of the entire review article. Write a concise summary of the entire review article. Write a concise summary of the entire review article.

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The abstract should include:

- Background/context of the selected research topic
- Purpose of the review
- General approach used to select and analyse the research articles
- Major findings from the reviewed literature
- Key research gaps identified
- Overall conclusion

Recommended length: 150–250 words.

Keywords: Keyword 1; Keyword 2; Keyword 3; Keyword 4; Keyword 5

Contents

1 Introduction

1.1 Background

As digital technology has experienced rapid evolution and expansion throughout the years, it has slowly integrated itself into most aspects of our own lives, from communications to our governance. Naturally, this boon that provides us convenience and connectivity will bring upon malicious actors that will misuse them to commit cybercrimes, ranging from small scaled crimes like identity theft and cybertheft of financial and card payment data to much larger scale ones such as cyberespionage and ransomware attacks.

As such, traditional methods of investigation have been proven to be ineffective in combating and solving these crimes. Hence, digital forensics as a discipline has seen a rise within the fields of cybersecurity and law enforcement to ensure that the malicious actors will have their deserved justice dealt to them alongside learning their methods to better suppress the surge of cybercrime.

Introduce the selected research topic and explain why it is important.

This is how you add inline references in the document given that all your references have been copied in the references.bib. Recent studies have demonstrated the effectiveness of artificial intelligence techniques (**smith2024**).

Follow this example to include more than one references. Several studies have investigated these approaches (**smith2024**; **lee2023**; **ahmad2022**).

Discuss:

- What is the research topic?
- What problem does it address?
- Why is the problem important?
- Where is the topic currently being applied?
- Why is there a need to study/review this topic?

Students should support important statements with appropriate research references.

1.2 Problem Statement

Almost every part of daily life now runs through a computer or a phone, and crime has followed right along with it. Money gets stolen, systems get broken into, and all of it can happen from the other side of the world. Digital forensics — the work of digging through devices and networks for proof of what happened — has become central to how law enforcement operates. That said, the field is under real strain right now.

For a long time, the basic strategy was straightforward: keep a list of known bad signatures and block anything that matched. That worked fine when attackers were unsophisticated. They aren't anymore. Advanced Persistent Threats, for example, are designed to slip past defenses and remain undiscovered for lengthy periods of time, silently removing important data from a network without raising any alarms. The scope of the problem is difficult to overestimate – cybercrime is expected to cost the global economy

10.5 trillion dollars by 2025, yet discovery is sometimes excruciatingly slow: one research estimated that the average time to find a breach was 212 days.

Part of the difficulty stems from the fact that digital forensics is a collection of disparate fields. Network forensics works with traffic, mobile forensics with phones and tablets, and then there's cloud, database, IoT, and multimedia forensics, each with its own set of tools and standards. A case that involves more than one of these areas requires investigators to piece together methodologies that were never intended to function together, slowing things slowly and introducing opportunity for mistake.

Criminals aren't making this easier, either. Encryption scrambles communications. Steganography hides malicious code inside what looks like an ordinary image. Fileless virus operates totally in memory and leaves no trace when the system reboots. Some players are even using blockchain to hide their actions. Even when investigators uncover anything, it does not always make it to court – shoddy paperwork, uneven processes, and contradicting legislation across jurisdictions are major reasons solid evidence is tossed out.

Artificial intelligence is frequently presented as the solution. It's easy to see why: machine learning is genuinely good at finding patterns buried in massive datasets and flagging behavior that a human analyst might miss, and there's evidence that AI-powered tools outperform older methods in terms of detection rate and false positives. But AI brings its own baggage. These models are frequently black boxes, so it's not always clear why a system flagged what it flagged. They can be manipulated with carefully designed adversarial inputs. They require massive training datasets, which are not always accessible, as well as significant computational capacity to function.

Then there's the smart-device problem. Homes are full of smart speakers, locks, thermostats, TVs; factories run thousands of interconnected sensors. All of this results in a massive, jumbled volume of data in inconsistent forms, dispersed among devices, cloud servers, and local storage. Figuring out where the relevant evidence even lives, let alone extracting it and proving it hasn't been tampered with, is genuinely difficult in this kind of distributed environment.

The legal landscape doesn't help. Privacy laws vary greatly by country; some countries openly share evidence while others do not, and permissible investigation tactics range from one another. Investigators must operate within this patchwork while adhering to guidelines such as GDPR and CCPA, and a misstep here might torpedo a case completely.

Despite years of research in this area, there is still no cohesive method that tackles all of these issues simultaneously. Standardized testing techniques for forensic tools are sparse, multi-domain datasets are scarce, and technologies produced by various manufacturers frequently do not work together. Furthermore, new dangers such as deepfakes and adversarial assaults continue to emerge and necessitate their own solutions.

1.3 Objectives of the Review

State the objectives of the review article.

This study aims to assess where digital forensics is today: what has been solved, what has not, and where the profession appears to be headed — with a focus on how AI and similar technologies might boost investigations and protect evidence integrity.

More specifically:

First, we look at current practice in six domains: network, mobile, database, IoT, cloud, and multimedia forensics, highlighting the key issues, the tools and methodologies

used, and the largest gaps in each.

Second, we evaluate how well AI-based methods function in practice, focusing on deep learning, federated learning, and anomaly detection to identify where these approaches succeed and where they fail .

Third, we examine the problems provided by new threats such as APTs, encryption, anti-forensic tactics, steganography, and AI-generated media, as well as potential responses .

Fourth, we look at the legal and ethical frameworks that shape the field, such as how courts assess the admissibility of digital evidence, chain-of-custody requirements, privacy protections, and the challenges that come with cross-border investigations.

Fifth, we present a cross-domain framework that combines federated learning with context-aware policy enforcement, which is intended to function across hybrid cloud, IoT, and corporate contexts while addressing the constraints of current single-domain techniques.

Finally, we highlight areas for future research, including explainable AI, quantum-resistant encryption, blockchain-based evidence tracking, privacy-preserving threat intelligence sharing, and automated attack attribution.

1.4 Research Questions

Develop 2–4 research questions that guide the review.

For Example:

RQ1: What approaches have been used to address [research problem]?

RQ2: What are the major findings reported in existing studies?

RQ3: What are the limitations of the existing approaches?

RQ4: What research gaps and future opportunities can be identified?

2 Review Methodology

Explain how the group conducted the literature review.

2.1 Literature Search Strategy

Describe:

- Databases/search engines used
- Search keywords
- Search strings
- Publication period
- Type of publications considered

Example databases:

- Scopus
- Web of Science
- IEEE Xplore
- ScienceDirect
- SpringerLink
- ACM Digital Library
- Google Scholar

2.2 Inclusion and Exclusion Criteria

Describe the criteria used to determine which research articles were included in or excluded from the review.

Inclusion criteria may include:

- Articles directly related to the selected research topic
- Peer-reviewed research articles
- Articles published within the selected publication period
- Articles indexed in the selected academic databases
- Articles available in full text

Exclusion criteria may include:

- Articles outside the scope of the research topic
- Duplicate publications
- Non-peer-reviewed publications
- Articles outside the selected publication period
- Articles without sufficient methodological or research information

Table 1: Inclusion and exclusion criteria used for the literature review

Criteria	Inclusion	Exclusion
Publication type	Peer-reviewed research articles	Non-peer-reviewed articles
Publication period	[e.g., 2021–2026]	Articles outside selected period
Language	English	Other languages
Topic	Directly related to selected topic	Not directly related
Full text	Full-text available	Full text unavailable

This is a sample table. Tables should be placed in the main text near to the first time they are cited, with the caption on top and left aligned. The font of the table caption is size 9 and the content in the table is size 10.

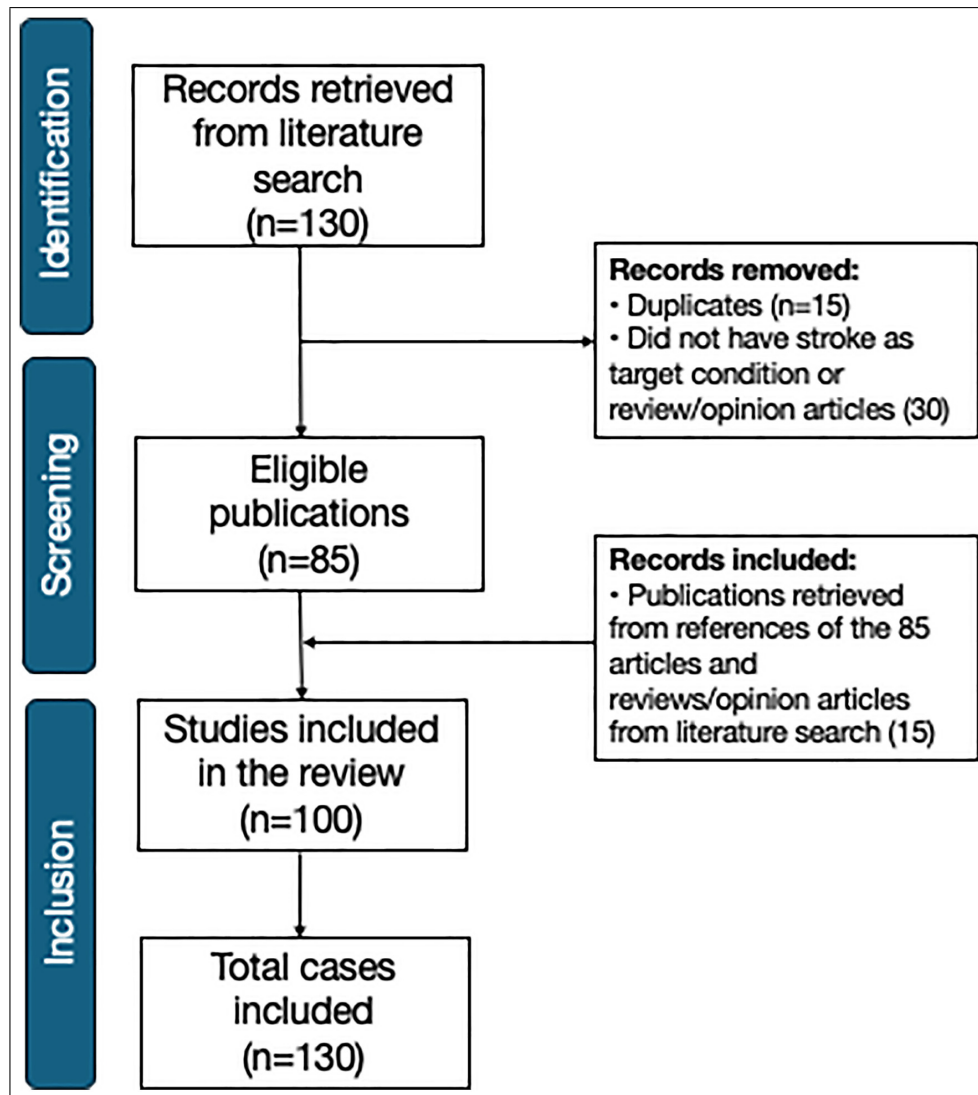


Figure 1: PRISMA flow diagram of the study selection process.

This is a figure. All figures should be centred with figure caption following the sequence Figure 1, Figure 2, and so on. The caption should also be centred. The caption for figures should be size 9.

2.3 Article Selection

Each student must review **at least 3 research articles**.

Therefore:

- **Minimum articles per student = 3**
- **Minimum articles per group = 12**

The group may use more than 12 articles if necessary.

Explain how the final articles were selected for the review.

3 Literature Review

3.1 Theme 1: Standardised Practices and Advancements of Digital Forensics

Through the rampant evolution of cybercrime in its sophistication of execution alongside the current digital age allowing access for various methods to execute cybercrime, digital forensics as a field within computer science has gone through many advancements to ensure that cybercriminals will face their due justice.

To better evaluate Digital Forensic's role in investigating and solving cybercrime, it would be apt to first understand the standardised methods of which the practitioners of digital forensics utilise in order to obtain the proper digital evidence for the committed cybercrime, and then delve into the various advancements made towards improving digital forensics' own practices as a whole.

Table 2 showcases research articles which align with the current theme of Standardised Practices of Digital Forensics, alongside their methodologies of research.

Table 2: Research Articles for Standardised Practices of Digital Forensics and their Methodology

Research Articles	Primary Objectives	Methodology
ilyas2024role	To analyze the extent and legal strength of digital forensic evidence within the Indonesian legal system.	Normative: Viewing law as a system of norms. Utilises philosophical (Nature of digital forensics), conceptual (Test and analyse concepts), and case (study and analyze electronic crime cases) approaches.
hassan2025	To evaluate the effectiveness of current digital forensic strategies and the possibility of enhancing them	Qualitative: Comparative analysis of digital forensic tools alongside examination of high-profile cybercrime case studies.
sharma2024digital	To investigate the adherence to legal standards and reporting procedures in forensic investigations.	Quantitative Content Analysis: Literature review of 40 reports from a quasi-experiment
ARIF2025104296	To evaluate the challenges of digital forensics the existing methodologies face	Systematic Literature Review (SLR): Analysis of peer-reviewed publications from 2019 to 2025 across prominent databases.

3.1.1 Comparison of Research Objectives and Scope

The chosen research articles intends to explore the transformation of digital forensics from traditional reactive models to much more reactive and intelligent frameworks.

- **General Evolution and AI Integration :** Research by **vaddi2024enhancements** focuses on the evolution of digital forensics over the past five years, specifically aiming to provide a high-level overview of AI's potential to solve modern difficulties. Their objective is to guide developers in constructing intelligent, automated tools for effective crime detection across blockchain, cloud, and deep learning (DL) domains **vaddi2024enhancements**.

- **Smart Environments and Forensic Readiness:** **alenezi2023digital** shifts the focus toward "smart environments" (homes, cities, and IoT). The primary objective is to survey recent advancements and challenges in retrieving and preserving evidence across a multitude of connected devices, while proposing potential solutions to overcome the lack of standardization in these environments (**alenezi2023digital**).

Table 3 showcases research articles which align with the theme of the Advancements of Digital Forensics, alongside their methodologies of research.

Table 3: Research Articles for Advancements of Digital Forensics and their Methodology

Feature	vaddi2024enhancements	alenezi2023digital
Research Type	Passive data gathering, literature and statistical analysis	Comprehensive survey
Temporal Focus	Patterns no older than 2018	Recent advancements and possible challenges
Core Methodology	Evaluating long-standing and latest techniques alongside summarising sections to keywords	Examining strengths and limitations of state-of-the-art tools and identifying major challenges and difficulties.

3.1.2 Standardised Investigative Procedures and Workflows

A central theme across the literature is the necessity of a structured, multi-stage forensic process to ensure the integrity and admissibility of evidence. While terminology varies slightly, the core stages remain consistent.

3.1.3 The Forensic Lifecycle Comparison

The Forensic Lifecycle Comparison Standardised practices generally adhere to a flow of identifying, securing, and interpreting data.

- **ilyas2024role** outlines a four stage model: Identification, Examination, Analysis and Reporting. The article emphasises the digital evidence obtained through the four stage model to be volatile and must be stored carefully for proper presentation of the evidence in a trial to ensure it is not defective in the eyes of the law.
- **hassan2025** instead presents a six-stage model: Identification, Collection, Preservation, Examination, Analysis, and Reporting. Key difference in comparison to the four stage model being the Collection and Preservation stages to ensure the integrity of the collected information, through use of write blockers, regular integrity checks and storage in secure, tamper proof containers
- **ARIF2025104296** instead suggests a smoother model, consisting of Identification, Preservation, Analysis, Documentation, and Presentation. Their focus is on a high-level theoretical metamodel designed to unify tasks across subdomains like IoT and Cloud forensics to reduce "process redundancies."

3.1.4 Key Procedural Requirements

Specific principles of digital forensics are treated as standardised practice, notably those cited from the Association of Chief Police Officers (ACPO) **ilyas2024role** :

- Non-interference: Digital data is prohibited to be changed before being brought to court.
- Competence: Anyone accessing digital evidence must show a degree of competence alongside understanding of the implications and the relevancy of their actions upon said evidence.
- Documentation: Steps taken on the storage media while its being analysed and investigated must be audited detailedly so that they are able to be replicated exactly with the same results by a third party.

3.1.5 Dataset, Tools, and Technology Comparison

To be able to acquire proper digital evidence, a multitude of tools will be required. One particular research article, particularly **hassan2025** manages to compare and contrast multiple various software and hardware solutions, all utilised in different ways both professionally and academically, which is properly displayed in Table 4.

Table 4: Forensic Tools and Use Cases

Category	Tools Identified	Strengths	Limitations
Computer Forensics	EnCase, FTK (Forensic Toolkit)	Comprehensive file systems and metadata analysis	High tool costs, high demand
Mobile Forensics	Cellebrite UFED, MSAB XRY, Oxygen Forensic Suite	Effective data extraction from locked/encrypted mobile devices	Remote-wiping of media, device damage
Network Forensics	Wireshark, NetWitness, Snort, Tcpdump	Real-time traffic monitoring and packet-level analysis	High expertise required, steep learning curve
Cloud/IoT	Magnet AXIOM, IoT Inspector	Specialized for multi-tenant environments and smart device vulnerabilities	Jurisdictional, privacy protocol and ethical concerns

3.1.6 Comparative Analysis of Technical Frameworks and Tools and their Advancements

The articles highlights a transition toward automated, decentralized, and layered forensic architectures.

1. Artificial Intelligence and Machine Learning Both articles agree that AI and ML are essential for managing the volume and complexity of data generated in modern environments.
 - **Architectural Components:** **vaddi2024enhancements** propose a hybrid model consisting of a User Interface (for crime type selection and data entry), a Feature Extraction Module (using OCR and voice recognition), and a Crime Matching Engine. This engine utilizes multi-linear perceptrons, K-nearest neighbor (KNN), and neural networks **vaddi2024enhancements**.
 - **Functional Applications:** **alenezi2023digital** categorizes AI/ML applications into image/audio analysis (facial/voice recognition), text analysis (sentiment and keyword identification in chat logs), and behavioral analysis (identifying suspicious patterns in user activity).
2. Cloud Forensics

- **Layered Frameworks:** **vaddi2024enhancements** describes a cloud framework separated into four distinct layers: abstract, information gathering, data storage, and information fusion. This framework utilizes a "voting-k-means" approach for evidence separation **vaddi2024enhancements**.
- **Investigation Focus:** **alenezi2023digital** emphasizes the analysis of virtual machine images and the reconstruction of user activity within cloud-based services, highlighting the need for cloud forensic readiness to address data ownership and privacy (**alenezi2023digital**).

3. Blockchain and IoT Integrity

- **Chain of Custody:** **vaddi2024enhancements** present a forensic-chain framework for IoT. This utilizes a distributed ledger to record analysis events (Technical Evidence or TEs), ensuring immutability through Merkle trees and simplified Proof of Work (PoW) algorithms **vaddi2024enhancements**.
- **Evidence Management:** The blockchain framework uses smart contracts to automatically log every evidence transfer, including timestamps, permission levels, and recipient addresses **vaddi2024enhancements**.

4. Network and Memory Forensics **alenezi2023digital** provides a detailed breakdown of network tools not extensively covered in the AI-centric **vaddi2024enhancements** study:

Packet Capture: Use of Wireshark, Snort, and tcpdump to identify suspicious traffic flows (**alenezi2023digital**).

- **Flow Analysis:** NetFlow and sFlow for detecting DoS and DDoS attacks.
- **Memory Forensics:** Analysis of computer memory to uncover malware and network intrusions in IoT and mobile devices (**alenezi2023digital**).

3.1.7 Experimental Setup and Data Sources

The research articles utilizes diverse data points to validate their own discovery and findings:

- **ilyas2024role:** Used data from the Indonesian National Police Headquarters (e-MP), analyzing 8,831 cybercrime cases prosecuted in 2022.
- **sharma2024digital:** Conducted content analysis on 40 reports generated via quasi-experiments to assess "Strength of Support" (SoS) categories.
- **hassan2025:** Analyzed high-profile case studies including the 2014 Sony Pictures Hack, the 2021 Colonial Pipeline Ransomware Attack, and the 2018 Cambridge Analytical Data Scandal

In terms of the research articles focused on the advancements of digital forensics however prioritises simulation and pattern recognition over specific, localized datasets.

- **Data Sources:** **vaddi2024enhancements** emphasize the importance of diverse internet resources and training datasets for more accurate and precise findings. They specifically mention the use of unstructured crime report narratives and past records from external agencies **vaddi2024enhancements**.

- **Visual Data:** Deep learning research focuses on forensic video analysis, specifically addressing low-quality CCTV footage. Techniques are devised to maximize the extraction of evidentiary elements through video enhancement **vaddi2024enhancements**.
- **Heterogeneity:** **alenezi2023digital** notes that the dataset in smart environments is inherently heterogeneous, consisting of data from sensors, social media, and location-based services, which complicates correlation and reconstruction (**alenezi2023digital**).

3.1.8 Results

The results of the research articles does reveals a discrepancy between theoretical standards and practical application:

- **Limitations of Reports:** **sharma2024digital** found that 82.5% of digital forensic studies were "Vague" in their methodology, and 0% explicitly asserted "Reliability" and "Validity" in a way consistent with other forensic sciences. This suggests a significant gap in academic and professional reporting standards.
- **Legal Admissibility:** **ilyas2024role** concluded that while digital forensics is critical for finding "material truth," its success in Indonesia is hampered by a limited number of experts and internal police regulations that do not fully support the complexity of contemporary cybercrime.
- **Automation and AI:** Arif et al. (2025) and **hassan2025** both noted a rising trend in the use of AI and Machine Learning to automate evidence classification and anomaly detection, which helps in managing the "explosion of technology" and data volumes.

On the other hand, Table 5 showcases the resultant findings in regards to the possible advancements in terms of utilisation of Artificial Intelligence and Machine Learning:

Table 5: Comparative Results and Findings for AI and ML

Category	vaddi2024enhancements	alenezi2023digital
Detection Accuracy	High efficacy in predicting illegal behavior and risk detection	AI/ML improves the speed and accuracy of analyzing large, complex datasets
Data Integrity	Effectively prevents evidence loss or corruption in insecure systems (Blockchain)	Forensic readiness policies reduce the time and cost associated with incident response
Data Integrity	Effectively prevents evidence loss or corruption in insecure systems (Blockchain)	Forensic readiness policies reduce the time and cost associated with incident response
Video/Media	Deep learning allows for extraction of evidence from low-quality footage	AI, through training is able to recognize specific suspects via voice or facial patterns

3.1.9 Advantages and Limitations of Current Practices

The research is able to identify the benefits of standardised practices alongside the advancements of digital forensics while acknowledging significant barriers to universal implementation as well as the flaws of said advancement of digital forensics.

In regards to the Standardised Practices of Digital Forensics, the benefits and the flaws of its implementation are as follows:

■ Advantages

- **Consistency:** Frameworks like the NIST guidelines provide a "consistent method of data retrieval" **hassan2025**).
- **Legal Integrity:** Adherence to chain-of-custody protocols ensures that evidence remains "legally solid" and admissible in court **ilyas2024role**.
- **Efficiency:** Standardised tools like Magnet AXIOM allow for cross-platform recovery (Cloud, Mobile, Computer), reducing the need for disparate systems.

■ Limitations

- **Encryption and Obfuscation:** Advanced encryption methods act as a primary barrier, preventing access to data despite standard protocols **hassan2025**).
- **Jurisdictional Complexity:** Cybercrime is inherently global, and the lack of a harmonized legal frameworks makes cross-border evidence sharing difficult (**ARIF2025104296; sharma2024digital**).
- **Rapid Obsolescence:** The speed of technological change often outpaces the development of forensic standards, particularly in the IoT and cloud sectors **hassan2025**).
- **Vague Reports:** As identified by **sharma2024digital**, a lack of specific methodological reporting hinders the ability of practitioners to replicate findings or rely on published research in a legal context.

In regards to the Advancements of Digital Forensics, the benefits and the flaws of its implementation are as follows:

■ Advantages

- **Automation:** Both studies highlight that AI and ML automate time-consuming manual processes, such as the visual scrutiny of video footage or the parsing of massive log files **alenezi2023digital**.
- **Transparency and Security:** The use of blockchain provides decentralization, removing the vulnerability of a "central authority" that might be targeted by attackers **vaddi2024enhancements**.
- **Proactive Readiness:** Forensic readiness planning allows organizations to collect evidence in a "forensically sound manner" before an incident occurs (**alenezi2023digital**).

■ Limitations

- **Technical Hurdles:** **vaddi2024enhancements** acknowledge the impossibility of an error-free framework, noting the persistence of false positives. **alenezi2023digital** highlights the challenges posed by encryption and the lack of standardization in communication protocols across different devices.

- **Ethical and Legal Constraints:** Both sources point to significant challenges regarding data privacy. **alenezi2023digital** specifically notes the difficulty in balancing the collection of personal data against the need to prosecute cybercrimes.
- **Resource Intensity:** Implementing these advanced frameworks (e.g., establishing a digital forensic laboratory or maintaining a blockchain network) requires substantial material and human resources (**alenezi2023digital; vaddi2024enhancements**).

3.2 Theme 2: AI Augmented Digital Forensics

The rapid evolution of cybercrime has necessitated a shift from traditional manual forensic techniques to AI-augmented digital forensics. As digital evidence proliferates across cloud platforms, encrypted services, and IoT devices, the volume and velocity of data have exceeded human processing capacities.

Through our gathered research articles, they highlight the integration of Artificial Intelligence (AI) not as a replacement for human judgment, but as a critical instrument for enhancing the speed, accuracy, and depth of cybersecurity investigations. (**sanna2025improving; 11385975**).

3.2.1 Comparative Analysis of Research Focus and Methodologies

The provided research context explores AI's role in digital forensics (DF) through different lenses, ranging from theoretical procedural frameworks to specific algorithmic testing and defense against sophisticated threats.

3.2.2 Similarities in Research Perspectives

Across the studies, several consensus begin to emerge:

- **The Scalability Crisis:** Researchers agree that traditional forensic methods is incapable of keeping up with the terabytes of logs and chaotic evidence produced during enterprise breaches **11385975**.
- **Human-Centric AI:** AI is framed as an "investigative partner" or "digital assistant" rather than a "black box" solution. Human supervision is deemed essential to validate findings and address the legal requirement for interpretability in court (**sanna2025improving; 11385975**).
- **The Adversarial Nature of AI:** There is a shared recognition in that cyber-criminals are leveraging AI for offensive purposes as well, such as automated intrusion, phishing, and anti-forensics techniques like steganography (**11166697; sanna2025improving**).

3.2.3 Methodological Differences and Research Goals

The researchers employ distinct strategies to address the advancement of the field through utilisation of AI, as demonstrated in Table 6.

Table 6: Comparative Results and Findings for AI and ML

Feature	sanna2025improving	11385975	1116666
Primary Focus	Procedural integration across the DF pipeline (4 Phase Procedure)	Technical performance of hybrid deep learning model	Defense against Persistent Threats
Research Methodology	Qualitative framework proposal and LLM case studies	Quantitative testing using CNN-BiLSTM on synthetic datasets	Comprehensive literature and APT case studies
Key Entities	General cybercrimes, LLMs (Gemini, Copilot, ChatGPT)	Ransomware, DDoS, and Insider Threats	State-sponsored Zero-day exploits
Core Argument	AI should be applied at all phases of digital forensics (From collection to reporting)	Hybrid models (CNN+Bi-LSTM) with attention are superior to standard ML	APTs require a multi-layered defense (Zero Trust)

3.2.4 AI Applications in the Digital Forensic Pipeline

The integration of AI is analyzed relative to the standard four-phase forensic procedure defined by NIST: Collection, Examination, Analysis, and Reporting.

1. Collection and Preservation

AI is able to facilitate the identification of digital evidence at crime scenes, through use of Computer Vision (CV) algorithms that can be deployed locally to assist first responders in identifying all relevant hardware (e.g., devices with volatile memory) to ensure all possible evidence has been recovered. AI also assists in terms of documenting the chain of custody by automatically recording and logging the device's conditions (**sanna2025improving**).

2. Examination and Data Acquisition

In the examination phase, AI serves as a digital assistant by guiding consultants through technical procedures, based on reviewing similar past cases. This is particularly useful for recovering data from corrupted memory sections or damaged disks. (**sanna2025improving**).

3. Analysis and Evidence Interpretation

This phase sees the most significant AI augmentation. Advanced models are used to:

- **Classify Multimedia:** Automated identification of specific data types (e.g., images, audio, chats) enhances privacy by only analyzing relevant data while searching for evidence. Besides that, it helps reduce potential trauma gain from analysts viewing sensitive and explicit materials (**sanna2025improving**).
- **Pattern Recognition:** Deep learning models, specifically hybrid architectures like CNN-BiLSTM, excel at capturing both spatial dependencies in digital artifacts and temporal sequences in network traffic **11385975**.
- **Anti-Forensics Detection:** AI can be trained to detect hidden patterns in files used for steganography or to identify AI-generated "deepfake" content that mimics real individuals (**sanna2025improving**).

4. Reporting and Legal Documentation Large Language Models (LLMs) and Natural Language Processing (NLP) are utilized to draft summaries and background technical definitions for legal parties. Future applications would include using NLP to review and score reports based on their understandability for non-technical stakeholders (**sanna2025improving**).

3.2.5 Technical Performance and Model Robustness

Empirical testing suggests that advanced AI architectures significantly outperform traditional machine learning models in forensic tasks, as proven from Table 7 which uses **11385975** research of utilising a hybrid CNN-BiLSTM model with an attention mechanism and comparing it against baseline models such as Logistic Regression and Random Forest.

Table 7: Comparison of Model Metrics

Model Type	Accuracy/F1-Score	Time-to-Discovery	False Positiv
Logistic Regression	0.78	High (12+ seconds)	12.5%
Random Forest	0.82	Moderate (9 seconds)	9.8%
Standard LSTM	0.86	Moderate (7 seconds)	7.4%
CNN-BiLSTM + Attention	0.93 - 0.96 (AUC)	Low (4 seconds)	3.1%

3.2.6 Strengths of the Hybrid Approach

The CNN-BiLSTM model’s success is attributed to its **Attention Mechanism**, which assigns higher weight to important evidence segments. Ablation testing confirmed that removing attention decreased precision of the model, while removing metadata embeddings significantly decreased the model’s recall **11385975**.

3.2.7 Challenges and Contradictory Findings

While AI offers substantial benefits, the article accurately identifies a few critical limitations and areas of friction within the implementation:

- The LLM Steganography Paradox **sanna2025improving** conducted a case study on popular LLMs (Gemini, Copilot, ChatGPT) regarding steganography.
 - Failure in Creation: None of the LLMs could successfully generate a working PNG image with steganographed content.
 - Success in Scripting: Conversely, the LLMs were highly proficient at generating functional Python code for encoding and decoding messages using Least Significant Bit (LSB) techniques.
 - Insight: This suggests LLMs are currently more useful as an offensive or defensive coding assistants rather than as direct content generators for anti-forensics.
- Attribution Complexity in APTs APTs (Advanced Persistent Threats) present a unique challenge due to their stealthy and state-sponsored nature. Attribution remains difficult because malicious actors use custom malware and fileless techniques

that runs only within the system memory instead of any other vulnerable areas. While AI improves threat hunting, the **automated attribution** of these actors remains a primary research gap (**11166697**).

- **Data Bias and Environmental Variability** A significant weakness in AI forensic tools is the risk of biased models. Models that are trained on Western corporate data may risk failure when applied to mobile-centric systems in developing countries. Ensuring fairness and accuracy requires diverse, globally-sourced training datasets/**11385975**.

3.2.8 Emerging Trends and Future Directions

While reading through the articles, a trend can be observed in the fact that digital forensics is moving towards more resilient, transparent and collaborative ecosystems.

- **Explainable AI (XAI)**: There appears to be a necessity for XAI to be properly utilised to provide clear justifications for the AI's own decisions. This transparency is vital for establishing trust among investigators and ensuring that the obtained evidence is admissible in court (**11166697**; **11385975**).
- **Blockchain Integration**: Researchers are exploring blockchain as a much more secure method for identity management, tamper-proof logging of forensic procedures alongside providing a decentralized threat intelligence sharing platform. (**11166697**).
- **AI-vs-AI Feedback Loops**: A proposed trend involves a continuous loop where offensive AI develops evasive techniques to better test defensive AI's responses. Human supervisors are required to monitor this interaction to better refine both of the AI's detection algorithms and analyst skills (**sanna2025improving**).
- **Federated Learning**: This technique allows organizations to collaboratively train APT detection models without sharing sensitive private data, thereby preserving privacy while increasing the shared knowledge base (**11166697**).

3.3 Theme 3: Cybercrime and the Detection of Digital Deceptions

3.3.1 Overview of the Cybercrime Landscape

The global cost of cybercrime is projected to reach \$10.5 trillion by 2025 **11480014**. This financial escalation is driven by the increasing sophistication of digital deception, primarily through two vectors: the democratization of high-fidelity synthetic media (deepfakes) and the refinement of traditional phishing via spoofed communications. Both methods exploit human trust and systemic vulnerabilities in standard authentication protocols.

3.3.2 Comparative Analysis of Deception Techniques

1. Deepfake Generation and Proliferation

According to **SINGH2025103632**, deepfakes utilize Generative Adversarial Networks (GANs) to create hyper-realistic images, audio, and video through techniques like face synthesis (e.g., StyleGAN), face morphing, and speech cloning

(e.g., Wav2Lip). This could be explained through the fact that open-source tools like DeepFaceLab and Faceswap have lowered the barrier of entry, allowing non-technical actors to generate deceptive content **SINGH2025103632**. No doubt the high prevalence of deepfakes like the Deepfake cases in India alone rose by 550%, with 90% of surveyed individuals in the UK expressing apprehension about the technology **SINGH2025103632**.

2. Email Spoofing and Phishing

11480014 highlight that phishing remains a dominant threat, with a 1726% increase in phishing websites between 2011 and 2021. Cybercriminals are able to use tools like malicious IP addresses, forged email headers, and tracking pixels which are used to monitor the recipient’s activity and location as an invasion of their privacy.

Unlike deepfakes, which often rely on visual/auditory realism, email spoofing instead leverages social engineering triggers such as emergencies and sudden authority presence (e.g., ”Microsoft account unusual sign-in activity”) to bypass the victim’s critical thinking **11480014**.

3.3.3 Detection Methodologies

To better combat the cybercrimes, researchers have propose diverse approaches to identifying digital deceptions, ranging from automated AI models to manual forensic analysis, as can be seen in Table 8.

Table 8: Detection Techniques and Capabilities

Technique	Application	Capabilities	Limitations
CNN-Based Models	Deepfakes (Image/Video)	High accuracy in spatial artifacts SINGH2025103632	Sensitive to compression artifacts
RNN/LSTM Models	Deepfakes (Image/Video)	Captures temporal inconsistencies and motion patterns SINGH2025103632	High computational cost
Header Analysis	Email Spoofing	Identifies mismatches in SPF, DKIM, and DMARC 11480014	Can be bypassed by sophisticated attackers
Biometric Defenses	Identity Verification	Liveness detection (eye blinks, pupil dilation) and 3D depth sensing SINGH2025103632	Subject-dependent performance
Hybrid Approaches	Multi-modal Detection	Combines human-in-the-loop verification with AI-based filtering 11480014	Complex design and resource-heavy

1. Similarities in Research Findings Both studies emphasize that **authentication failures** are primary indicators of fraud. In email forensics, this manifests as **spf=none** or **dkim=none** results **11480014**. In deepfake detection, this involves identifying ”spoofing” where synthetic data is presented as genuine biometric data **SINGH2025103632**. Furthermore, both identify **demographic and environmental biases** in training data (e.g., FFHQ and DFDC datasets) as a significant hurdle for universal detection **SINGH2025103632**.

2. Differences and Contradictory Findings

- Scalability: **SINGH2025103632** note that heavy AI pipelines (Transformers) are computationally expensive and struggle with real-time requirements (running at 18–22 fps). Conversely, **11480014** argue for the continued necessity of manual forensic analysis, asserting that automated tools often overlook subtle anomalies like a 2-second timestamp mismatch in email hops.
- Human Factor: **11480014** emphasises human behavior as a **critical vulnerability**, whereas **SINGH2025103632** focuses more on the technical **arms race** between generative models and detection algorithms while arguing for the human element for the detector models.

3.3.4 Strengths, Weaknesses, and Vulnerabilities

1. Technical Weaknesses

- **Cross-Dataset Generalization:** Models trained on specific datasets (e.g., DFDC) often fail when tested against "in-the-wild" content. For example, the accuracy of the models can drop from 90% to 60% when facing user-generated content like compressed TikTok videos **SINGH2025103632**.
- **Adversarial Attacks:** Attackers can add **intentional perturbations** (noise) to forged media to deceive CNN-based classifiers into labeling them as authentic **SINGH2025103632**.

2. Forensic Failures Real-world incidents demonstrate the stakes of these vulnerabilities:

- **Political Turmoil:** The 2019 Ali Bongo incident in Gabon showed how a suspected deepfake (even if genuine) can trigger military takeover attempts due to public mistrust **SINGH2025103632**.
- **Financial Loss:** A 2019 voice deepfake attack against a UK-based CEO resulted in a \$243,000 fraudulent wire transfer, illustrating the danger of voice cloning in high-stakes environments **SINGH2025103632**.

3.3.5 Emerging Trends and Policy Frameworks

- Research Trends While analysing the research articles for this review, certain trends tend to pop up. One in particular being the Explainable AI (XAI), which are models that provide reasonings alongside an interpretable output to support legal admissibility and public trust in their usage **SINGH2025103632**. Another one would be the facilitation of Federated Learning, a privacy-preserving approach where models are able to be trained across devices without centralizing raw biometric data **SINGH2025103632**. In addition, there seems to be talks of the implementation of continuous authentication: a system that monitor user behaviors (gait, keystroke rhythm) throughout a whole session rather than relying on a one-time login **SINGH2025103632**, which could better ensure the user isn't a malicious actor that pretends to be another user.
- Global Regulatory Initiatives In terms of global efforts in the other hand, it appears that there are new legal frameworks being established, with the aim to categorize and mitigate AI risks. For one, the EU AI Act (2024) has categorised deepfakes

as **high-risk**, requiring transparency measures such as labeling synthetic content **SINGH2025103632** to avoid misconception. As observable from NIST’s Incident Response, **11480014** advocates for integrating the NIST Incident Response Life-cycle with current forensic investigations to standardize how organizations should handle spoofing attacks. In addition, a brand new strategy has appeared: **Provenance Tagging**, which uses cryptographic hashes to track the origin of digital media, is currently being supported by the Partnership on AI (2023) to identify tampered content **SINGH2025103632**.

4 Comparative Analysis of Existing Studies

Provide a systematic comparison of the reviewed research articles.

A table such as the following is recommended.

Table 9: Comparative analysis of the reviewed research studies

No.	Author & Year	Research Objective	Method/Technique	Dataset/Sample	Key Findings	Limitations
1	[Author, Year]	[Research objective]	[Method/Technique]	[Dataset/Sample]	[Key findings]	[Limitations]
2	[Author, Year]	[Research objective]	[Method/Technique]	[Dataset/Sample]	[Key findings]	[Limitations]
3	[Author, Year]	[Research objective]	[Method/Technique]	[Dataset/Sample]	[Key findings]	[Limitations]
4	[Author, Year]	[Research objective]	[Method/Technique]	[Dataset/Sample]	[Key findings]	[Limitations]

5 Synthesis of Findings

This section is particularly important.

The group should answer:

5.1 What Do the Studies Agree On?

Identify findings or conclusions that are consistently reported across multiple studies.

5.2 What Do the Studies Disagree On?

Identify conflicting findings, different results, or different conclusions.

5.3 Which Approaches Appear to Be Most Effective?

Compare the approaches and explain why certain methods appear to perform better.

5.4 What Trends Can Be Observed?

Identify emerging trends in:

- Technologies
- Methodologies
- Datasets
- Algorithms

- Applications
- Research directions

The discussion should be based on evidence from the reviewed articles.

6 Research Gaps

Identify gaps that remain in the existing literature.

Possible research gaps include:

- Lack of suitable datasets
- Small sample sizes
- Limited geographical coverage
- Poor generalisation
- Limited experimental validation
- Lack of comparison between methods
- High computational requirements
- Lack of real-world testing
- Security/privacy concerns
- Lack of explainability
- Outdated technologies
- Limited integration with other technologies

For each identified gap, provide evidence from the reviewed literature.

7 Future Research Directions

Based on the identified research gaps, propose potential directions for future research.

For example:

1. Development of improved algorithms
2. Use of larger datasets
3. Integration of emerging technologies
4. Development of real-time systems
5. Improved accuracy/performance
6. Cross-domain applications

7. Improved security and privacy
8. Explainable AI approaches
9. Real-world deployment and validation

The proposed directions should logically follow from the research gaps identified in Section ??.

8 Discussion

Provide an overall critical discussion of the literature.

The discussion should answer:

- What has the research community achieved so far?
- Which approaches are most promising?
- What are the major challenges?
- Are the current approaches sufficient?
- What limitations remain?
- How has the research area evolved?
- What important issues remain unresolved?

This section should demonstrate **critical thinking**, not merely summarisation.

9 Conclusion

Summarise the major findings of the review.

The conclusion should:

- Restate the purpose of the review
- Summarise the major findings
- Highlight important research gaps
- State the overall conclusion
- Briefly indicate future research opportunities

Do not introduce completely new information or references in the conclusion.

APPENDIX A – INDIVIDUAL ARTICLE REVIEW

Each student should complete the following table for their **minimum 3 research articles** before the group combines the findings.

Student: [Name / Student ID]

Table 10: Individual article review

Article	Research Problem	Objective	Methodology	Dataset/Sample	Key Findings	Limitations	Research Gap

This appendix provides evidence of each student’s individual contribution to the group review.

APPENDIX B – ARTICLE MATRIX

The group should combine all individual article reviews into one matrix.

Table 11: Group article matrix of reviewed research studies

Article	Student	Year	Method	Dataset	Key Results	Strength	Limitation	Research Gap
1	Student 1							
2	Student 1							
3	Student 1							
4	Student 2							
5	Student 2							
6	Student 2							
...	...							